

DAF26BX06-NV510 - AI/ML for Next Generation of Missile Detection, Warning, Tracking, and Reporting

USAF / SpaceWERX with SSC Space Sensing Delta 84 and the OPIR Tap Lab | SBIR | Release 6 CSO | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE (feasibility-study track). The TPOC confirmed in Q&A that a simulation-driven trade study with no new detection algorithm is responsive. Small award, short period, ITAR staffing constraint. Confidence MEDIUM-HIGH.

Phase I award	\$150,000, 3 months (SpaceWERX CSO)
Proposal format	Technical volume limit 30 pages (per the DAF CSO index table)
Compliance	ITAR/EAR clause: foreign nationals must be disclosed and may be restricted. CMMC Level 2 (Self).
Phase II	Development and testing inside the OPIR Tap Lab or similar testbed; transition path is SSC's Space Modernization Initiative
Deliverable	Feasibility study + CONOPS + analytical findings; algorithm prototypes or simulation results 'as applicable'

What they want

[TOPIC TEXT] "Proposals should emphasize technical feasibility and clearly outline a Phase I research plan that can include modeling, simulation, literature review, or trade studies... Phase I deliverables are expected to focus on a feasibility study detailing USSF CONOPS and technical viability, as well as analytical findings."

Four focus areas, any part of which is acceptable: edge fusion and autonomy (on-orbit track fusion, constellation tasking); multi-source integration (OPIR + EO + radar + SIGINT + commercial); OODA-loop collapse and mission assurance (orchestration, edge computing, 'AI trade-space exploration', explainability); and new sources/methods outside current TTPs.

The feasibility study must identify benefits over existing solutions, risks with mitigations, critical technology elements, and evidence for each CTE (literature, trade studies, analysis, heritage).

What the Q&A changed

- [Q&A] A "deterministic simulation that screens the multi-orbit OPIR sensing trade space... to find where AI/ML fusion would have real leverage before Phase II" - "Yes, a feasibility study would be appropriate for this topic."
- [Q&A] An in-space edge-compute architecture effort (hosting, not inventing, the algorithms) is responsive; document it with DoDAF 2.0-style architecture products.
- [Q&A] "Mission-aware orchestration and AI-assisted trade-space decision management" is "an appropriate submission" for the OODA focus area.
- [Q&A] Future use of the Phase I product (offline tool vs. Tap Lab plug-in) is decided at the end of Phase I.

How we would build it

- **Constellation and sensing simulator:** propagate synthetic GEO/MEO/LEO constellations (public two-line-element style orbits), model sensor fields of view, revisit and data cadence, and link latency.
- **Threat set:** synthetic boost/glide trajectories for ballistic and maneuvering profiles from open-literature kinematics - no real threat data.

- **Fusion placement experiments:** compare ground-centralized vs. on-orbit/edge fusion for warn-to-decision latency, track continuity through handovers, and robustness to dropped links; identify where AI/ML data association adds value versus classical filters.
- **Explainability/confidence:** propagate confidence through the pipeline so the CONOPS can state what an operator sees and when automation hands off.
- **Deliverables:** the feasibility study (benefits, risks, CTEs with evidence), a system-level CONOPS, and the simulation as a reusable analysis tool with a stated path into the OPIR Tap Lab.

Resources we would use

- Public orbit propagation tooling (SGP4-class libraries, open astrodynamics packages) and public catalog data for constellation geometry.
- Open literature on OPIR missile-warning architectures (Next-Gen OPIR, HBTSS-class concepts as publicly described) and multi-sensor tracking (JPDA/MHT, learned data association).
- OPIR Tap Lab public material (boulderlab.org) for integration expectations.
- GAO TRL guide (cited in the topic) for the maturity language reviewers expect.
- No hardware. Compute is modest.

Bottlenecks and risks

- **ITAR staffing:** any non-US-person contributor must be disclosed and may be excluded from technical data.
- **Three months, \$150k:** scope must be a crisp trade study, not a platform; over-promising algorithms is the classic failure here.
- **Domain vocabulary:** missile-warning CONOPS reviewers expect fluency (revisit, dwell, handover, typing). A space-sensing advisor would materially improve the CONOPS section.
- **Incumbency:** established OPIR ground-processing primes are around this program; the Tap Lab exists precisely to onboard non-trationals, which helps, but Phase II integration will require working inside their environment.
- **Transition dependence:** value realization depends on being invited into the Tap Lab; Phase I is a foot in the door.

Decision factors

- Fit to team: simulation + data pipelines + ML - direct; astrodynamics is learnable at the fidelity a trade study needs.
- Dual use: [INFERRED] multi-orbit sensor tasking and fusion resemble commercial EO-constellation scheduling and space traffic management; the topic itself names none.
- Strategic value: opens the SpaceWERX/SSC relationship at low cost; the 3-month award is small but fast.
- Decisive pre-proposal experiment: build a two-constellation revisit/latency simulator against a synthetic launch and show one chart where fusion placement changes warn-to-decision time - that chart belongs in the proposal.

Before Oct 7 (Q&A): ask whether unclassified OPIR-like sample data or Tap Lab sandbox access is available in Phase I, and what architecture artifacts the Tap Lab expects at handoff.

Tags: [TOPIC TEXT] = quoted from the solicitation; [Q&A] = government answer on DSIP; [INFERRED] = our judgment, not verified by search. Generated 2026-09-20 from release6_triage.md / triage_results.md / sitis_qa.md.